Course Name: ETHICAL HACKING

Assignment- Week 3

TYPE OF QUESTION: MCQ/MSQ/SA

Number of questions: 10 **Total mark: 10 x 1 = 10**

QUESTION 1:

Which of the following statement(s) is/are true regarding connectionless routing used by the IP protocol?

a. Each packet is treated as an independent unit for routing decisions.
b. A fixed path is established before any packet is transmitted.
c. Different packets between the same source and destination may follow different paths.
d. IP protocol uses connection-oriented routing.

Correct Answer: a, c

Detailed Solution: IP uses connectionless routing, where each packet is treated independently and may follow a different path even between the same source-destination pair. IP does not establish a fixed path beforehand (that is connection-oriented behavior).

Thus, the correct options are (a) and (c).

QUESTION 2:

Which of the following statement(s) is/are **false** regarding direct and indirect packet delivery?

a. Direct delivery occurs when the destination host and deliverer are on the same physical network.
b. Indirect delivery occurs when the destination host is not on the same network as the deliverer.
c. In direct delivery, packets always pass through at least one router.
d. In indirect delivery, the packet is routed hop by hop until it reaches a router on the destination's network.
e. Direct delivery can occur between the last router and the destination host.

Correct Answer: c

Detailed Solution: Direct delivery occurs when source and destination share the same physical network, or between the last router and the destination host — hence it does NOT always require passing through a router, making (c) false. All other statements correctly describe direct/indirect delivery.
Thus, the correct (false) option is (c).

QUESTION 3:

Which routing table flag indicates that a route is specific to a single host rather than an entire network?

a. U
b. G
c. H
d. D
e. M

Correct Answer: c

Detailed Solution: The H flag in a routing table entry indicates that the route is host-specific (a route to a single host), as opposed to a route to an entire network.
Thus the correct option is (c).

QUESTION 4:

In the OSPF routing protocol, which type of packet is used to determine whether a neighboring router is active?

a. Link State Update
b. Link State Request
c. Hello packet
d. Link State Acknowledgement
e. TCP SYN packet

Correct Answer: c

Detailed Solution: OSPF uses "Hello" packets periodically exchanged between neighbouring routers to check whether a neighbour is alive/active. If Hello packets are not received within a timeout period, the neighbour is considered down.
Thus, the correct option is (c).

QUESTION 5:

Which type of address should be used when a packet needs to be delivered to every host within a specific network?

a. Unicast address
b. Broadcast address
c. Anycast address
d. None of these

Correct Answer: b

Detailed Solution: A broadcast address is used to deliver a packet to all hosts within a given network or subnetwork. Unicast targets a single specific host, and anycast targets the nearest of a group of hosts.

Thus, the correct option is (b).

QUESTION 6:

Which of the following statement(s) is/are **false** regarding routing protocols?

a. RIP uses distance vector routing and can suffer from the count-to-infinity problem.
b. OSPF is a link-state routing protocol.
c. BGP is used for routing within a single autonomous system.
d. OSPF uses Dijkstra's algorithm to compute shortest paths.
e. RIP is typically used for routing between different autonomous systems.

Correct Answer: c, e

Detailed Solution: BGP is an exterior routing protocol used for routing between different autonomous systems, not within a single AS, making (c) false. RIP is an interior protocol used within a single AS, not between different ASes, making (e) false. RIP's distance vector approach can indeed suffer from count-to-infinity, and OSPF is a link-state protocol using Dijkstra's algorithm - both correctly stated.

Thus, the correct (false) options are (c) and (e).

QUESTION 7:

What is the address used to represent the default route in a routing table (in IPv4)?

a. 127.0.0.1
b. 255.255.255.255
c. 0.0.0.0
d. 192.0.0.1
e. None of these

Correct Answer: c

Detailed Solution: The default route, also called the "gateway of last resort," is represented by the address 0.0.0.0 (with subnet mask 0.0.0.0, or /0 in CIDR), and is used for packets that don't match any other entry in the routing table.

Thus, the correct option is (c).

QUESTION 8:

Which of the following statement(s) is/are **false** regarding IPv6?

a. IPv6 supports address classes similar to IPv4 (Class A, B, C).
b. IPv6 addresses are typically represented in hexadecimal, separated by colons.
c. IPv6 eliminates the need for fragmentation by intermediate routers.
d. The base header size of IPv6 is variable, ranging from 20 to 60 bytes.

Correct Answer: a, d

Detailed Solution: IPv6 does not use classful addressing like IPv4 (no Class A/B/C), making (a) false. The IPv6 base header has a fixed size of 40 bytes (not variable 20-60 bytes, which describes IPv4), making (d) false. IPv6 addresses are indeed hexadecimal and colon-separated, fragmentation in IPv6 is done only by the source (not intermediate routers).

Thus, the correct (false) options are (a) and (d).

QUESTION 9:

In IPv6 base header _____ bits are used for Hop Limit?

Correct Answer: 8

Detailed Solution: The Hop Limit field in the IPv6 base header is 8 bits wide, serving a similar function to the TTL field in IPv4 - it is decremented at each hop, and the packet is discarded when it reaches 0.

Thus, the correct answer is 8.

QUESTION 10:

An organization is assigned 172.20.64.0/20 network block. _____ number of valid host addresses are available in this block.

Correct Answer: 4094

Detail Solution: A /20 prefix leaves 12 host bits, giving 2^12 = 4096 addresses. Subtracting the network address and broadcast address leaves 4094 usable host addresses.

**********END*******